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Privatization's Unequal Toll: Explaining Cross-Country Variation in Mortality and Health Outcomes After Transition From Communism

Master's degree thesis

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Summary

The thesis covers the health impact of large-scale privatization during the post-communist transition (1989-2014), which triggered divergent demographic trajectories across 14 economies. Two-Way Fixed Effects and Staggered Event Study models isolate the causal impact of rapid property transfers on mortality. Rapid mass privatization without institutional buffers contributed to a severe crisis, reducing average life expectancy by 2.36 years. This shock was profoundly gendered, with working-age men suffering a 2.69-year penalty driven by acute psychosocial stress. Conversely, robust state capacity and active civil society participation effectively buffered these negative health consequences, mitigating the transition-induced cardiovascular trauma.

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privatization, mortality, transition economics

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Scientific paper

Introduction

The collapse of communism in Central and Eastern Europe as well as in the former Soviet Union after 1989 when the Berlin Wall fell, marks a significant turning point in global politics and economics. The complexity of carrying out politico-economic regime transformation simultaneously in a variety of institutional systems is often referred to as the "dilemma of synchronicity" (Ramet, 2010). This concept underscores the immense challenge faced by the states in attempting to democratize political systems, redefine state borders or identities, and restructure economies concurrently, rather than sequentially. In the case of transition to free markets from centrally-planned economies in the Soviet Bloc (Soviet Union and aligned countries), a cornerstone of the regime shift was privatization, which entailed the introduction of free-market principles (Estrin et al., 2004). This process targeted a diverse range of state-owned assets, including both enterprises established as public property and those that had been previously privately owned before being nationalized (Ramet, 2010). Furthermore, the execution of privatization varied significantly across the region; while some countries adopted rapid "shock" or mass voucher schemes, others pursued a more gradualist approach favoring direct sales or restitution (Soós, 1997; Roháč, 2016). Privatisation was expected to result in increased efficiency and economic growth, as well as improved living standards and quality of life (Cornia, 2016b). These expectations were set against a stark baseline, as the preceding decades under the communist regime were marked by stagnating life expectancies and high mortality levels, largely driven by chronic negligence, technological backwardness, and the profound systemic inefficiencies of the state-run health care systems (Eberstadt, 1981; Feshbach and Friendly, 1992; Meslé and Vallin, 2002). However, the outcomes of the transition have been far from uniform across countries, with some countries experiencing significant improvements in health and mortality rates, while others have faced deteriorating conditions (Brainerd, 1998).

Present paper aims at investigating the specific mechanisms behind this demographic divergence, focusing on the velocity of large-scale privatization. Specifically, it seeks to determine whether the rapid transfer of state assets to private markets directly contributed to the post-socialist mortality crisis, how this shock disproportionately affected different genders, and to what extent institutional factors—such as state capacity and civil society participation—acted as protective buffers. By analyzing a cross-country panel dataset of 14 post-communist economies between 1989 and 2014, this study bridges the gap between macroeconomic policy choices and distinct epidemiological outcomes, including cardiovascular disease and external causes of death.

Specifically, we posit that the rapid implementation of large-scale privatization directly contributed to the post-socialist mortality crisis. To understand the diverging outcomes across the region, we further hypothesize that high state capacity—operationalized through government expenditure—mitigated these mortality-inducing effects by acting as a structural buffer. Finally, we examine whether strong civil society participation served as a social capital shield, cushioning the negative health consequences by providing informal support networks that reduced transition-induced psychosocial stress.

The present study contributes to the literature by reconciling the debate between shock therapy and gradualist approaches, demonstrating non-linear privatisation effects. The main findings reveal that rapid large-scale privatization without institutional buffers caused a direct decline in overall life expectancy of 2.36 years in the region. Crucially, the analysis uncovers a profound gender asymmetry: working-age men bore the absolute brunt of the shock, suffering a penalty of 2.69 years compared to a 1.74-year decline for women. Furthermore, the specific implementation of mass voucher privatization in the Former Soviet Union carried a catastrophic penalty of 2.55 years. Conversely, strong state capacity, through government expenditure, and robust social capital effectively mitigated this mortality shock, particularly reducing deaths from cardiovascular and external causes among men.

Furthermore, advanced robustness checks confirm the highly specific and contextual nature of this crisis. By disaggregating the transition strategy into a categorical shock index, the analysis reveals that while mass privatization in Central Europe yielded no statistically significant demographic harm, the exact same policy mechanism executed within the "institutional void" of the Former Soviet Union drove the demographic collapse. This demonstrates that the "Unequal Toll" of the transition was not an inherent consequence of moving towards private ownership, but a direct result of chaotic policy execution in an unbuffered environment.

The remainder of the paper is structured as follows: the next section reviews the existing literature on transition types and academic debate on the association between privatization and mortality. The subsequent section details the data and methodology, including the Staggered Two-Way Fixed Effects estimation strategy. Finally, the empirical results and robustness checks are presented, followed by the concluding remarks.

1. Relevant Literature

1.1. Transition Types

The widespread institutional collapse across Central and Eastern Europe and the Former Soviet Union in the early 1990s led to an imperative for simultaneous political, territorial and economic restructuring. Existing studies in the academic literature seek to define and categorize these complex processes, resulting in multiple analytical approaches (Bujwid-Kurek and Mikucka-Wójtowicz, 2015). In this study we refer to the seminal typology of transformation processes (Bujwid-Kurek and Mikucka-Wójtowicz, 2015) by Samuel Huntington that divides countries into three categories based on the nature of their reforms (Huntington, 1991):

- "transformation" (e.g. Soviet Union, Bulgaria, Hungary) when the group of reformators originates in communist government elites,
- "transplacement" (e.g. Czechoslovakia, Poland) when the reforms are imposed by both communist government elites and opposition forces, as it happened during the round table talks,
- "replacement" (e.g. East Germany, Romania) when the opposition to the communist government has a significant role in reforms.

These categories reflect the degree of domestic consensus and external pressure influencing the reform processes. Furthermore, it is important to note that these transition processes were complicated by internal factions. The ruling elite typically split into reformers and hardliners, while the opposition was divided into moderates and radicals. In case of other classifications, they mostly match the above-mentioned typology, although some differences in countries grouping can be observed. Sometimes, a separate category is being distinguished for the dissolution of multi-national federations, such as Czechoslovakia or Yugoslavia, due to the combined effect of elite-level political fragmentation and grassroots nationalistic pressure (Wiatr, 2020; Bunce, 1999). It is worth mentioning also the classification proposed by Kitschelt (2001) due its emphasis on the group of countries where preemptive reforms were the key idea. Examples are Soviet Union Gorbachev's initial innovations, as well as the regime changes in Bulgaria, Romania etc.

In sum, despite substantial similiarities in specific types of transition in country groups, there is major heterogeneity not only across but within these country groups. The country-specific variation in the regime transition yields the opportunity to examine its elements' impact on the outcomes relevant to the aim of present study, namely mortality.

1.2. Economic Transformation and the Privatization Debate

Privatization was universally deemed a central component of the economic transformation, necessary to replace the inefficient command economy with market principles (Estrin, 1994; Berg and Sachs, 1992). While domestic political contexts varied across the region, the broader policy debate regarding the pace and method of implementation was dominated by a dichotomy between two camps. The first camp consisted of neoliberal advisors who championed "shock therapy" advocating for the rapid and simultaneous implementation of price and trade liberalization, stabilization and mass privatization (Hamm et al., 2012). This approach was intended to create an irreversible shift to the market. In contrary, critics often described as gradualist economists or neo-institutionalists, argued that hasty reform damages the state and hence, they favored a slower, more gradual process to allow sufficient time for essential governing institutions (e.g. corporate law, capital markets) to develop (Roháč, 2016; Stuckler et al., 2009).

To accurately examine these processes, it is crucial to make a clear distinction between the broader concept of transition—understood as the comprehensive political, social, and institutional shift away from communism (Ramet, 2010)—and privatization, which was a specific economic mechanism within that transformation (Estrin, 1994; Berg and Sachs, 1992). Furthermore, the subject of this privatization encompassed a vast array of state-owned assets (Ramet, 2010), ranging from large-scale industrial enterprises to small-scale properties, including real estate, retail shops, and housing (European Bank for Reconstruction and Development, 1994, 1999). The specific privatization methods chosen across Central and Eastern Europe and the Former Soviet Union yielded varied results and attracted distinct academic critiques. Regarding mass (voucher) privatization, while empirical results suggest that it generally had a significantly positive effect on growth after 1995, following the initial recession (Estrin et al., 2004), its implementation faced scrutiny. Particularly in the Czech Republic, this method was criticized for leading to "budgetary crises and unleashing inflation at the macroeconomic level" (Marcinčin and van Wijnbergen, 1997). Conversely, insider privatization models that favored management and employees, such as those in the Russian Federation and Ukraine, resulted in "insider dominance" and were generally less effective than market methods (Estrin et al., 2004; Djankov and Murrell, 2002), fostering the development of crony capitalism in the longer run (Hoff and Stiglitz, 2004; Stiglitz, 1999). This approach was associated with political continuity and was mostly restricted in the Czech Republic and Hungary, and to a lesser extent in Poland, where a hard political transition had weakened the positions of former communist managers (Soós, 1997).

Overall, the literature confirms that the transition enhanced corruption due to the massive transfer of property coupled with weak state institutions and underdeveloped civil societies (Hoff and Stiglitz, 2004; Stiglitz, 1999; Kaufmann and Siegelbaum, 1997). Research indicates that privatization's overall effect on economic growth hinges on the quality of the institutional setting, often proxied by the level of corruption (Cieślik and Goczek, 2018). Furthermore, the implementation of mass privatization programs is documented to create a massive fiscal shock for post-communist governments, which undermined state capacity and the protection of property rights, thereby exacerbating the recession (Hamm et al., 2012). Ultimately, public dissatisfaction with privatization was driven by the flawed execution of the reforms and their adverse social outcomes, rather than an ideological opposition to private property itself (Denisova et al., 2012). Consequently, the popular demand for revising privatization was strongly correlated with individuals who experienced hardships during the transition, such as wage cuts, unemployment, or poor self-rated health.

Building on this, the literature suggests that the institutional setting acts as a critical moderator of privatization outcomes (Cieślik and Goczek, 2018; Denisova et al., 2009). While mass privatization can undermine state capacity (Hamm et al., 2012), the presence of robust governance and active civil society participation may serve as a buffer against the most disruptive social effects (Stuckler et al., 2009; Cornia, 2016a). This perspective shifts the focus from the mere pace of reform to the structural environment in which property transfer occurs (Roland, 2001), suggesting that countries with higher levels of civic engagement and better enterprise restructuring oversight might mitigate the transition's negative health consequences (Stuckler et al., 2009; European Bank for Reconstruction and Development, 1999).

1.3. Mortality and Health Outcomes

The political and economic upheaval in Central and Eastern Europe and the Former Soviet Union led to a severe post-socialist mortality crisis between 1989 and 1995, resulting in an estimated 10 million excess deaths between 1990 and 2000 (Cornia, 2016a). Importantly, there is a significant heterogenity across countries. Specifically, two general trajectories can be distinguished. First, the countries that faced a Severe Crisis (the Former Soviet Union states, including Russian Federation, Ukraine and Belarus) and experienced dramatic and prolonged mortality reversals (Eberstadt, 1999). Russian Federation male life expectancy, for example, dropped by 6.6 years between 1989 and 1994, reaching a level three years below that of India, which was then a poor agrarian economy (Cornia, 2016b). The death toll in post-Soviet states reached an estimated 7 million premature deaths, with 4 million in the Russian Federation alone (Azarova et al., 2018). Second, the trajectory of Rapid Recovery (Central European countries, such as the Czech Republic and Poland), were the first former Communist countries to see a "rapid and sustained increase in life expectancy" after 1991 (Fihel, 2011).

To understand the drivers of these demographic shifts, it is necessary to examine the specific structure of mortality. The literature identifies the crisis as being primarily fueled by a surge in cardiovascular diseases (particularly ischemic heart disease) and external causes (accidents, suicide and alcohol poisoning) (Cornia, 2016a; Azarova et al., 2018).

The excess mortality was concentrated among working-age men, though with distinct patterns across cohorts. The surge was especially pronounced among young adults (aged 20–39 years), who were predominantly affected by the external death causes, and older working-age adults (40–59), who suffered mainly from a rise in cardiovascular diseases leading to death (Cornia, 2016b). In the Russian Federation, for instance, the fastest relative mortality increases occurred in the younger group, while the largest absolute rise in mortality was recorded among the older cohort (Eberstadt, 1999). In attempting to explain this sudden surge, part of the literature attributes it to a "rebound effect" following the cessation of Mikhail Gorbachev's strict anti-alcohol campaign from the mid-1980s (Bhattacharya et al., 2013; Treisman, 2010; Leon et al., 1997). Nevertheless, recent re-examinations demonstrate that while alcohol policies and its availability were contributing factors, their impact fails to explain the crisis entirely, which suports the notion that privatization might have contributed to the increased mortality (Azarova et al., 2019).

The specific medical and health-related outcomes help to better understand the mechanisms underlying mortality changes. Various biological markers can serve as measures of acute psychosocial stress triggered by rapid economic dislocation and unemployment (Stuckler et al., 2009). These circumstances manifest precisely through the channels observed above: physiological strain leading to heart failure (Brainerd, 1998; Leon and Shkolnikov, 1998), and severe psychological distress that drives self-destructive coping behaviors - specifically rampant alcoholism and drug abuse (Zaridze et al., 2009) - which ultimately culminate in "deaths of despair" (King et al., 2009).

Despite the moderating role of institutions, a fundamental division persists in the literature regarding the extent to which mass privatization itself was the decisive upstream factor driving mortality crises. First, a large body of cross-national and quasi-experimental research supports the hypothesis that rapid and extensive privatization (often referred to as mass privatization) was a "crucial determinant" of the adverse mortality trends (Stuckler et al., 2009; Azarova et al., 2018, 2019; King et al., 2009). The primary mechanism linking this policy to premature death is identified as acute psychosocial stress caused by rapid economic dislocation. For example, one quasi-experimental retrospective cohort study in post-Soviet Russian Federation found that fast-privatized towns experienced 13% higher mortality among working-age men than slow-privatized towns (Azarova et al., 2018). These findings provide compelling evidence that rapid privatization contributed to raised working-age male mortality.

In case of female related research, results based on Hungarian town-level data show that prolonged state ownership was associated with a strong protective effect against mortality during the post-socialist transformation (Scheiring et al., 2018). Specifically, women residing in towns with sustained state-owned enterprises experienced significantly lower odds of dying compared to those in rapidly privatized areas. This was largely because state employment shielded these cohorts from abrupt economic dislocation, unemployment, and the resulting transition-induced psychosocial stress (Scheiring et al., 2018).

Conversely, other prominent studies challenge the conclusion that mass privatization was the primary cause of the crisis, citing methodological weaknesses and a lack of empirical robustness in the initial macro-level studies (Earle and Gehlbach, 2010; Gerry, 2012). Some researchers argue that the claim that mass privatization adversely affected male mortality trends “does not stand up to closer examination” and is contradicted by other knowledge about health trends in the region (Earle and Gehlbach, 2010). Gerry (2012) critiques the study by Stuckler et al. (2009) for committing an ecological fallacy, arguing that their broad, country-level analysis fails to prove that mass privatization was directly responsible for causing death at the individual level. Furthermore, Earle and Gehlbach (2013) demonstrate that the cross-country correlation between privatization and mortality disappears when accounting for pre-existing country-specific mortality trends and introducing short time-lags for policy implementation. Crucially, using firm-level data, they find no robust evidence that enterprise privatization systematically led to massive job losses, thereby questioning the underlying unemployment-stress mechanism.

To reconcile these conflicting narratives, recent scholarship has moved beyond cross-country aggregates to explicitly test the privatization thesis against competing theories. Systematic reviews of the literature have shown that while the debate remains active, four out of five cross-national studies found rapid privatization to be significantly and positively related to mortality (Scheiring et al., 2019). Moreover, to ensure that the privatization effect is not merely a statistical artifact of omitted variables—such as the dramatic changes in alcohol affordability (Treisman, 2010) or the Gorbachev campaign's rebound effect (Bhattacharya et al., 2013)—subsequent micro-level studies tested these hypotheses simultaneously. Azarova et al. (2019) confirm that the link between rapid privatization and increased mortality remains highly robust even when rigorously controlling for regional alcohol policy.

However, despite these compelling micro-level findings, the overall scientific knowledge on the effects of politico-economic transitions on health outcomes remains "inconclusive" and there is room for further examination, which the rest of the paper will be devoted to. Specifically, to address this gap, this study formulates and tests three main hypotheses:

- Hypothesis 1 (The "Shock" effect): The rapid implementation of large-scale privatization directly contributed to the post-socialist mortality crisis (Stuckler et al., 2009; King et al., 2009).
- Hypothesis 2 (The State Buffer): High state capacity, operationalized through government expenditure, mitigated the mortality-inducing effects of rapid privatization (Hamm et al., 2012).
- Hypothesis 3 (The Social Capital Shield): Strong civil society participation buffered the negative health consequences by providing informal support networks that reduced transition-induced psychosocial stress (Stuckler et al., 2009; Scheiring et al., 2018).

2. Identification Strategy

2.1. Main Analysis

To estimate the impact of structural transformation and privatization strategies on population health, the study employs as the baseline a Panel Ordinary Least Squares (Panel OLS) estimator. To transparently demonstrate the influence of both local unobserved factors and global time shocks, the econometric analysis follows a stepwise evolution. The modeling begins with one-way fixed-effects specifications that strictly control for unobserved, time-invariant heterogeneity across countries (μ_i), such as deep-seated cultural attitudes toward health or historical institutional legacies.

Subsequently, to rigorously eliminate confounding macroeconomic trends that affected all transition economies simultaneously (e.g., the 1998 financial crisis), the analysis is extended to a Two-Way Fixed Effects (TWFE) framework in the main analysis (Wooldridge, 2010; Allison, 2009).

The fully specified TWFE baseline equation used to test Hypothesis 1 (the "Shock" effect) is formally defined as follows:

$$H_{it} = \alpha + \beta_1 Priv_{it} + \gamma X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{1}$$

where H_{it} represents the specific health outcome (e.g., Life Expectancy or SDR from External Causes) for country i in year t. $Priv_{it}$ denotes the primary proxy for structural reform (Large-scale Privatization) and β_1 is the parameter of interest capturing the health effect of these reforms (Stuckler et al., 2009; Brainerd, 2001). X_{it} is a vector of time-varying covariates, including the natural logarithm of real GDP per capita, the employment rate, the inflation rate, average alcohol consumption, health equality, and a binary dummy for active armed conflicts (War). The terms μ_i and τ_t represent country and year fixed effects, respectively, while ε_{it} is the idiosyncratic error term.

To formally test Hypotheses 2 and 3, positing that state capacity and social capital acted as protective buffers (Hamm et al., 2012; Scheiring et al., 2018), the above TWFE model is extended with the interaction terms. Specifically, these buffering mechanisms are modeled as follows:

$$H_{it} = \alpha + \beta_1 Priv_{it} + \beta_2 (Priv_{it} \times Buffer_{it}) + \beta_3 Buffer_{it} + \gamma X_{it} + \mu_i + \tau_t + \varepsilon_{it} \tag{2}$$

In this specification, $Buffer_{it}$ represents either Government Expenditure (testing H2) or the Civil Society Participation Index (testing H3). The coefficient β_2 is the critical parameter for the “State Buffer” and “Social Capital” hypotheses, evaluating whether higher institutional capacity mitigated the mortality-inducing effects of rapid privatization.

Furthermore, to rigorously test the epidemiological stylized fact that the transition shock disproportionately targeted specific demographic groups, the primary TWFE specification is systematically disaggregated by gender. By estimating the models separately for male and female life expectancy, the identification strategy captures the profound gender asymmetry of the mortality crisis and mathematically evaluates whether male cohorts were significantly more reliant on institutional buffers than women.

Finally, to ensure the validity of statistical inference, all models are estimated using cluster-robust standard errors at the country level. This adjustment addresses potential issues of heteroskedasticity and within-country serial correlation, which are common in longitudinal macro-health data and could otherwise lead to over-optimistic p-values (Gerry, 2012; Cameron and Miller, 2015; Bertrand et al., 2004).

In addition, to explicitly isolate the policy mechanism of the crisis within the Former Soviet Union (FSU), the estimation strategy incorporates a time-invariant binary dummy for Mass Voucher Privatization. Estimated using Pooled OLS, the effect captures the specific mortality penalty of executing rapid property transfers without protective institutional buffers (Hamm et al., 2012).

The final model specification in the main analysis addresses the dynamic temporal effects of the transition and corrects for potential biases inherent in static TWFE estimators, such as the inability to distinguish between pre-trends and dynamic post-treatment effects. Specifically, we refer to a Staggered TWFE (Event Study) framework. We define the “treatment” threshold as the year a country’s Large-scale Privatization index reaches a critical transition phase (e.g., an EBRD score of 3.0 or higher). By establishing relative time indicators centered around the cut-off date, we estimate the following event study specification:

$$H_{it} = \alpha + \sum_{k=-3, k\neq -1}^{4} \delta_k D_{it}^k + \gamma X_{it} + \mu_i + \tau_t + \varepsilon_{it}$$

where D_{it}^k represents a set of binary dummy variables taking the value of 1 if year t is exactly k years away from the initiation of mass privatization for country i, and 0 otherwise.

The period immediately preceding the shock ($k = -1$) serves as the omitted reference category.

This dynamic specification serves a critical dual purpose in our identification strategy. First, examining the pre-treatment coefficients (δ_{-3}, δ_{-2}) validates the parallel trends assumption by confirming the absence of anticipating demographic shifts before the reforms. Second, the post-treatment coefficients ($\delta_0, \dots, \delta_4$) allow us to map the precise trajectory of the mortality penalty over time. This enables us to empirically determine whether the privatization shock induced a permanent structural decline in health or an acute, transient crisis that faded as institutions stabilized.

2.2. Robustness Checks and Analytical Software

To ensure the validity and stability of the baseline findings, several robustness checks are conducted. First, to mitigate potential concerns regarding reverse causality—where rapidly deteriorating population health might theoretically slow down the pace of economic restructuring—the core models are re-estimated using one-year lagged values of the EBRD transition indicators ($Priv_{it-1}$) (Gerry, 2012; Earle and Gehlbach, 2013; Wooldridge, 2010).

Second, to confirm the physiological specificity of the stress mechanism, a falsification (placebo) test is implemented using the Standardized Death Rate (SDR) from Malignant Neoplasms as the dependent variable (Leon et al., 1997; Brainerd and Cutler, 2005). Additionally, to verify that the mortality penalty was specifically tied to the period of peak institutional instability, a temporal interaction term isolating the 1990s Shock Therapy era is introduced. Furthermore, the study addresses potential omitted variable bias by evolving the specification from simple entity effects to a Two-Way Fixed Effects (TWFE) framework. This controls for unobserved global shocks (e.g., the 1998 Russian financial crisis (European Bank for Reconstruction and Development, 1999)) that could otherwise confound the relationship between privatization and mortality.

All data integration processes, feature engineering, and econometric estimations were performed using Python (Van Rossum and Drake, 2009). Specifically, the *linearmodels* library was utilized for high-dimensional panel data analysis (Sheppard, 2021), with data visualization and diagnostic plotting supported by *seaborn* and *matplotlib*. Crucially, while linear interpolation of institutional covariates was employed for the baseline models to preserve the core sample size, the formal falsification tests and temporal lag models are executed on a strictly restricted dataset. By explicitly dropping all interpolated variables and utilizing only raw macroeconomic indicators, the study guarantees that the documented privatization penalty is a robust epidemiological fact, not an artifact of missing data treatments.

3. Data and Measures

3.1. Data Sources and Sample Selection

To examine the health consequences of structural transformation, a longitudinal cross-country dataset was constructed, covering 14 post-communist economies in Central and Eastern Europe (CEE) and the Former Soviet Union (FSU) over the period 1989–2014. The sample includes Belarus, Bulgaria, Czech Republic, Estonia, Hungary, Kazakhstan, Latvia, Lithuania, Poland, Romania, Russian Federation, Slovakia, Slovenia, and Ukraine. This specific group was selected based on the completeness and historical continuity of cause-specific mortality records in international databases during the early transition period. The study population represents distinct transition trajectories, geographically and institutionally clustered into Central Europe (Poland, Czech Republic, Slovakia, Hungary), the Baltic states (Estonia, Latvia, Lithuania), South-Eastern Europe (Bulgaria, Romania, Slovenia), and the FSU (Russian Federation, Ukraine, Belarus, Kazakhstan). The temporal scope captures the immediate pre-transition baseline (1989), the turbulent early transition years characterized by transformational recession (Cornia, 2016b; Kornai, 1994), and the subsequent stabilization phases up to 2014.

The dataset integrates demographic and mortality data from the World Health Organization (WHO) European Health for All Database (HFA-DB) (World Health Organization Regional Office for Europe, 2026). To capture the economic and institutional dimensions of transition, structural reform indicators from the European Bank for Reconstruction and Development (EBRD) (European Bank for Reconstruction and Development, 2026) and historically reconstructed GDP series from the Maddison Project Database (Jutta Bolt and Jan Luiten van Zanden, 2020) were utilized. Complementary data on social conditions, institutional quality, and state capacity were obtained from the Varieties of Democracy (V-Dem) project (Coppedge et al., 2024), the World Bank's World Development Indicators (WDI) (The World Bank, 2026), and OECD Health Data (Organisation for Economic Co-operation and Development, 2026). Additional labor market and social expenditure statistics were sourced from UNICEF databases and reports (UNICEF Regional Office for CEE/CIS, 2013; United Nations Economic Commission for Europe, 2004). Supplemental macroeconomic controls (inflation, private sector share) utilize historical data from EBRD Transition Reports (European Bank for Reconstruction and Development, 1994, 1999, 2004). While these repositories are widely considered the gold standard for macro-level research, the reliability of early-1990s data - particularly within the FSU - constitutes a recognized limitation due to administrative collapse, which often resulted in systemic reporting inconsistencies (Stuckler et al., 2009; Gerry, 2012).

To address these inconsistencies and ensure robust estimation, the raw dataset underwent extensive cleaning and feature engineering. Historical series for employment and government expenditure were initially harmonized across multiple sources (HFA-DB, UNICEF, and TransMONEE). Structurally, the initial framework constitutes a balanced panel ($N = 14, T = 26$) comprising 364 country-year observations. However, a naive listwise deletion of missing values on this unharmonized dataset would drastically reduce the volume to merely 174 cases. To minimize this loss of valuable data points, missing values for institutional and macroeconomic covariates were linearly interpolated - strictly provided that the gaps did not exceed two consecutive years.

Following this rigorous data cleaning, which included the systematic recoding of non-numeric historical artifacts into valid numerics, the effective, high-quality working sample size for the main TWFE models were finalized at 169 observations spanning 12 countries. Because data missingness in this context is not entirely random - but rather heavily concentrated during the most chaotic early years of the transition shock (1989–1993) - relying exclusively on a strict complete-case analysis risks systematically dropping the most severe phases of the demographic crisis. The combination of strict data harmonization and restricted interpolation prevents this sample selection bias, avoiding the artificial smoothing of the mortality spike and yielding a more reliable representation of the transition's true toll.

3.2. Measures

Outcome Variables: Mortality and Health

Population health is operationalized using Standardized Death Rates (SDR) per 100,000 population (Brainerd, 1998; Stuckler et al., 2009). To disentangle the specific mechanisms linking economic shock to mortality, a broad spectrum of cause-specific outcomes available in the WHO HFA-DB is analyzed (World Health Organization Regional Office for Europe, 2026). To ensure cross-country comparability and longitudinal consistency across the 1989–2014 transition period, the causes of death are strictly defined according to the Ninth and Tenth Revisions of the International Classification of Diseases (ICD-9 and ICD-10). Utilizing both revisions is methodologically essential, as the timeframe of this study spans the global transition in national mortality reporting standards (World Health Organization, 1977, 1992).

First, to capture the physiological toll of the transition shock and directly test the stress mechanisms proposed in Hypotheses 1 and 3, the SDR from Ischaemic Heart Disease (defined by ICD-9 codes 410–414 and ICD-10 codes I20–I25) is employed as the primary biological marker (Cornia, 2016b; Leon and Shkolnikov, 1998). Second, to evaluate the specific prediction that social capital buffers against anomie and behavioral breakdown (Hypothesis 3), the SDR from External Causes of Injury and Poisoning (ICD-9 codes E800–E999 and ICD-10 codes V01–Y98), which inherently encompasses transport accidents, suicides and alcohol poisonings, is examined as a proxy for the resulting societal instability and psychological distress (Stuckler et al., 2009; Brainerd, 2001).

Furthermore, to move beyond aggregate trends and investigate specific behavioral channels, the analysis independently examines SDR from Suicide and SDR from Alcohol-related causes. By utilizing these as distinct dependent variables alongside the broader external causes category, the study can differentiate between acute physiological stress responses (Ischaemic Heart Disease) and specific "deaths of despair" resulting from long-term social maladaptation (Zaridze et al., 2009; King et al., 2009; Brainerd, 2001). This multi-layered approach allows for a more nuanced investigation of how institutional buffers, such as social capital and state expenditure, mitigate different forms of psychological distress and behavioral pathologies.

Consistent with rigorous methodological standards, the SDR from Malignant Neoplasms

(ICD-9 codes 140–208 and ICD-10 codes C00–C97) is employed as a negative control (placebo outcome). Since cancer mortality is driven by long-term etiological processes, it should theoretically remain unresponsive to short-term economic shocks (Leon et al., 1997; Brainerd and Cutler, 2005). Life Expectancy at Birth (LEB) is also reported as an aggregate measure of population welfare (Cornia, 2016b; Brainerd, 1998).

The SDR from Circulatory System diseases is included in the descriptive dataset to provide a comprehensive baseline of the cardiovascular crisis, although the more specific Ischaemic Heart Disease index is prioritized in the regression models for its higher sensitivity to acute structural shocks (Cornia, 2016b; Zaridze et al., 2009).

Finally, recognizing that the post-communist mortality crisis was distinctly male-centered (Brainerd, 1998; Leon and Shkolnikov, 1998), the demographic specificity of the transition shock is investigated by systematically disaggregating Life Expectancy and cause-specific mortality rates by gender. By estimating separate models for men and women, the analysis formally evaluates the gender-specific asymmetry of the privatization penalty and determines whether male cohorts were significantly more reliant on institutional buffers to survive the macroeconomic dislocation.

Explanatory Variables: The Multidimensional Transition Shock

To quantify the intensity of structural reforms, the study utilizes the complete set of transition indicators provided by the EBRD. While the primary focus lies on ownership transformation, the multidimensional nature of the market shock is accounted for by including the following indices:

- Large-scale Privatization: Progress in the transfer of large state assets to private hands.
- Small-scale Privatization: Progress in the privatization of small-scale assets.
- Governance and Enterprise Restructuring: Progress in hardening budget constraints and corporate governance.
- Price Liberalization: Progress in lifting price controls.

- Trade and Forex System: Liberalization of foreign trade and currency exchange.
- Competition Policy: Development of anti-monopoly legislation and institutions.

All EBRD indices are measured on a scale from 1.0 (centrally planned economy) to 4.3 (advanced market economy) (Stuckler et al., 2009; European Bank for Reconstruction and Development, 1999). However, a diagnostic correlation analysis of these indicators revealed a high degree of synchronicity in the reform process, with Pearson coefficients exceeding 0.80 for the majority of index pairs. To ensure model parsimony and mitigate the risk of severe multicollinearity, the study selects Large-scale Privatization (LSP) as the primary proxy for the structural transition shock (Gerry et al., 2010).

Literature further suggests that subjective indices may be prone to perception bias and fail to distinguish between different methods of property transfer (Hamm et al., 2012; Stuckler et al., 2009). Therefore, the Private Sector Share of GDP is included as an objective measure to validate the structural results (Hamm et al., 2012; Bennett et al., 2007). While the broader study population is conceptually categorized into four distinct sub-regions, the econometric estimation specifically employs a binary dummy variable to isolate mass privatization in the Former Soviet Union (FSU). This targeted dichotomy is theoretically motivated: it explicitly separates the radical, voucher-based schemes executed within an "institutional void" in the FSU from the more gradual or robustly institutionalized reform paths observed across Central Europe and the Baltics (Estrin et al., 2004; Earle and Gehlbach, 2013; Roland, 2000; King, 2001).

Covariates and Moderators

The model accounts for a rich set of time-varying confounders to isolate the effect of privatization from broader socio-economic shifts. Economic development is controlled for using the natural logarithm of *Real GDP per capita* (constant 2011 US$, Maddison Project), capturing the non-linear relationship between national wealth and population health (Hamm et al., 2012; Stuckler et al., 2009).

As a proxy for labor market security and economic inclusion—factors crucial during the industrial restructuring of the 1990s—the *Employment Rate* (sourced from UNICEF TransMONEE) is employed. This indicator offers superior data completeness and reliability compared to official unemployment statistics, which in the early transition period often failed to capture hidden unemployment and the collapse of the socialist "full employment" guarantee (Cornia, 2016b; United Nations Economic Commission for Europe, 2004). Macroeconomic instability, which can trigger acute stress through the erosion of savings and purchasing power, is controlled for using the annual *Inflation Rate* (Brainerd, 1998; Treisman, 2010).

To test the buffering role of formal and informal institutions, the *Civil Society Participation Index* (specifically, the V-Dem indicator *v2csprtcpt*) is incorporated as a proxy for social capital. This multidimensional indicator reflects the strength of autonomous civic groups, which theoretically mitigate anomie by providing social support networks (Stuckler et al., 2009; Scheiring et al., 2018). State capacity is captured by *Government Consumption Expenditure* (% of GDP), representing the fiscal "buffer" available to maintain public services (Hamm et al., 2012; Cieślik and Goczek, 2018). Additionally, the model accounts for *Health Equality* to measure the fairness of health protection systems (Cornia, 2016b).

Finally, the analysis controls for behavioral risks using average *Alcohol Consumption* (liters per capita) and adjusts for exogenous shocks through a binary variable for *War*, identifying years of active armed conflict (Stuckler et al., 2009; Bhattacharya et al., 2013).

Table 1 provides a summary of all variables included in the dataset.

Table 1: List of Variables and Data Sources

Variable Category	**Specific Indicators Included**
Health Outcomes (WHO)	Life Expectancy at Birth, SDR Ischaemic Heart Disease, SDR External Causes, SDR Suicide, SDR Alcohol-related Causes, SDR Malignant Neoplasms (Placebo).
Transition Indicators (EBRD)	Large-scale Privatization, Small-scale Privatization, Governance & Restructuring, Price Liberalization, Trade & Forex System, Competition Policy.
Economic Structure	Private Sector Share of GDP (EBRD), Employment Rate (UNICEF), Inflation Rate (EBRD).
Macro-Controls	Real GDP per Capita (Maddison Project), Government Consumption Expenditure (WB WDI).
Social & Institutional	Civil Society Participation (V-Dem), Health Equality (V-Dem), Alcohol Consumption (OECD), War (Dummy).

Source: Author's own elaboration based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

4. Results

4.1. Descriptive Statistics and Data Diagnostics

Before estimating the panel fixed-effects models, a preliminary analysis of the data distribution and potential multicollinearity was conducted. Table 2 presents the summary statistics for the principal variables used in the analysis.

Table 2: Descriptive Statistics (1989–2014)

Variable	Count	Mean	Std. Dev.	Min	Max
Life Expectancy	359	71.70	3.41	64.03	81.34
SDR External Causes	358	103.62	47.24	31.72	247.96
SDR Ischaemic Heart Disease	357	273.47	115.83	52.52	543.70
Large-scale Privatization (EBRD)	357	2.89	1.04	1.00	4.00
Mass Privatization (Dummy)	364	0.24	0.43	0.00	1.00
Government Expenditure (% GDP)	332	18.54	3.31	10.17	29.93
Civil Society Participation (V-Dem)	354	0.66	0.18	0.04	0.93
Real GDP Per Capita	364	14612.13	5750.79	4612.00	28474.00
Employment Rate (%)	342	52.41	7.12	34.20	68.40
Alcohol Consumption	305	10.20	2.17	3.30	14.80
Inflation Rate	210	167.22	497.09	-1.20	4735.00
War (Dummy)	364	0.06	0.24	0.00	1.00

Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

The sample reveals a profound degree of demographic and economic heterogeneity characteristic of the transition era. Life expectancy at birth averaged 71.7 years, but this aggregate figure masks a staggering gap of 17.3 years between the minimum (64.03 years in the early 1990s FSU) and the maximum (81.34 years) (Brainerd and Cutler, 2005). Similarly, mortality from Ischaemic Heart Disease (IHD) shows extreme volatility, with a maximum value of 543.70 per 100,000 population, underscoring the severity of the cardiovascular crisis in unbuffered regions (Zaridze et al., 2009; Brainerd, 2001).

Crucially, this demographic divergence is mirrored by the substantial variance in institutional protection across the cohorts. The indicators for Government Expenditure and Civil Society Participation show sufficient variation to test the buffering hypotheses, with civic engagement ranging from near-total absence (0.04) to high levels of participation (0.93). This wide spectrum of institutional strength – from the "institutional void" of the early-transition FSU to the more robust social-democratic frameworks in Central Europe – provides the necessary statistical leverage to evaluate whether state and social capital effectively moderated the mortality-inducing effects of rapid structural reforms (Cornia, 2016b; Hamm et al., 2012).

To ensure the robustness of the econometric estimations, Pearson correlation coefficients and Variance Inflation Factors (VIF) were examined. A primary concern in transition studies is the potential overlap between different reform paths and economic development. Indeed, the correlation between Large-scale and Small-scale privatization was exceptionally high (0.87), which strictly justified the selection of Large-scale Privatization as a single, primary proxy to avoid multicollinearity. Furthermore, formal multicollinearity tests confirmed that all VIF values for the selected base controls remained below 3.5 (with the highest being 3.19 for Log GDP). This indicates that the variables can be safely included in a single multivariate model without risking inflated standard errors.

4.2. Exploratory Data Analysis and Stylized Facts

Before delving into the formal econometric modeling, a visual exploration of the raw data provides critical context for the hypotheses. It is essential to establish the chronological baseline: the broader structural transition commenced with the geopolitical shifts of 1989 for Central and Eastern Europe (CEE) and the formal dissolution of the USSR in 1991 for the Former Soviet Union (FSU). Subsequently, the core policies of mass privatization were predominantly initiated and executed between 1992 and 1994. Figure 1 illustrates the divergent trajectories of life expectancy at birth across the three distinct privatization strategies following these pivotal years.

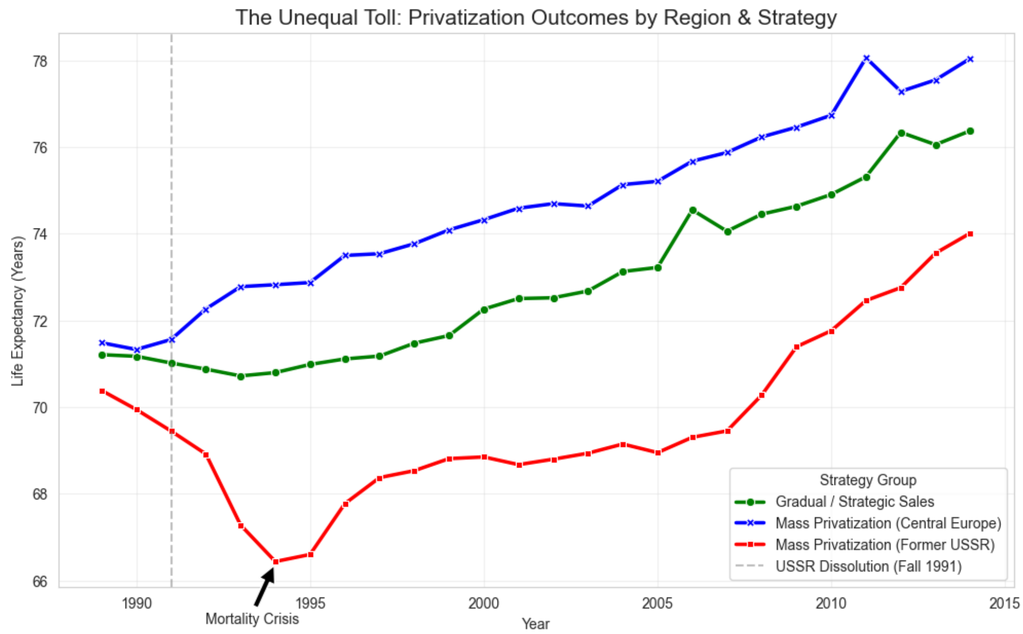

Figure 1: The Unequal Toll: Life Expectancy Outcomes by Region and Strategy

Source: Author's own elaboration based on WHO HFA-DB data.

The visual evidence reveals stark heterogeneity in post-communist health trajectories. While countries pursuing gradual reforms or mass privatization buffered by strong institutions (Central Europe) experienced only a brief stagnation followed by steady demographic recovery, the FSU cohort suffered a dramatic collapse exactly as the 1992–1994 property transfers took effect.

To further dissect this heterogeneity, Figure 2 presents the individual country trajectories, highlighting what this study terms the "Equal vs. Unequal Toll" phenomenon - a reflection of the sharply divergent post-communist demographic trajectories extensively documented in the foundational literature (Cornia, 2016b; Brainerd, 1998; Stuckler et al., 2009). This visualization reveals that while the initial 1989/1991 transition shock was universal, the demographic response was highly idiosyncratic. In the "Unequal Toll" group - exemplified by the Russian Federation, Ukraine, and Kazakhstan - we observe a sharp, non-linear collapse in life expectancy strictly coinciding with the peak privatization reform years (1992–1994). In contrast, "Equal Toll" countries like Poland and Slovenia maintained stable or improving health trends, suggesting that the mortality crisis was not an inevitable byproduct of privatization, but a consequence of the institutional environment.

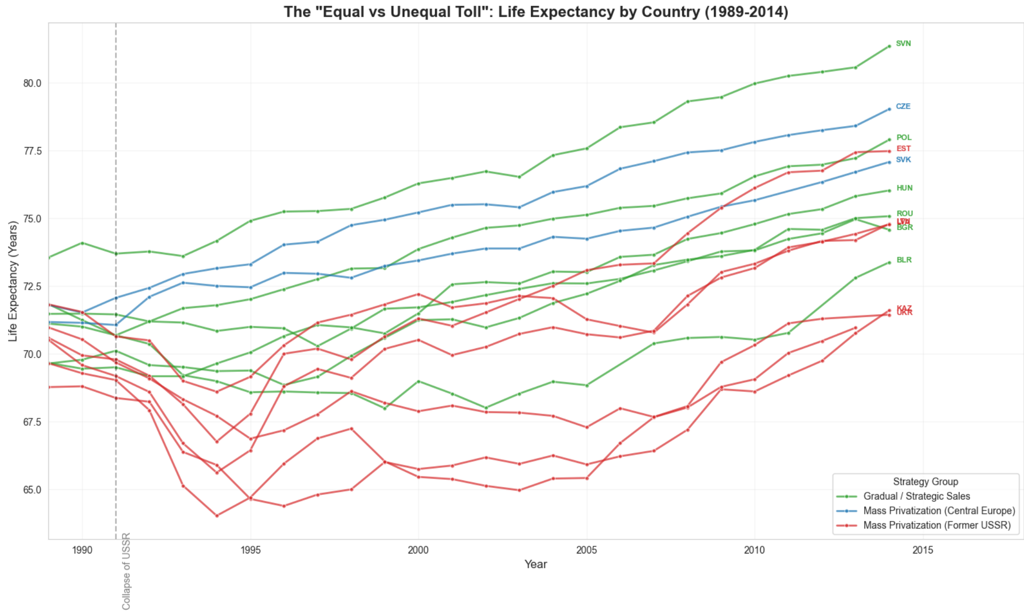

Figure 2: The "Equal vs. Unequal Toll": Life Expectancy by Country (1989–2014)

Source: Author's own elaboration based on WHO HFA-DB data.

To understand the policy-level drivers of this divergence, Figure 2 traces the pace of property restructuring using the EBRD Large-scale Privatization Index. The data clearly distinguishes the aggressive "shock therapy" implemented in the FSU and Central European mass privatizers from the incremental property transfers characterizing the gradualist approach.

However, the speed of ownership transfer alone does not fully explain the crisis; Central European states applied similar speed but avoided the health collapse. Figure 3 highlights the concept of the "institutional void" by plotting the scale of privatization against the Civil Society Participation Index during the critical mid-1990s period.

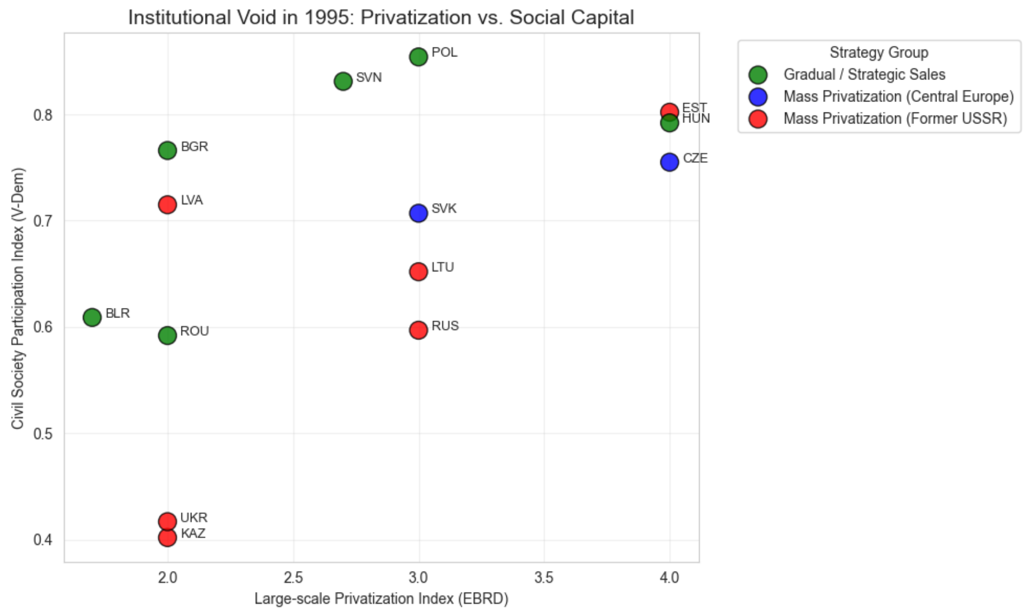

Figure 3: Institutional Void in 1995: Privatization Scale vs. Social Capital

Source: Author's own elaboration based on EBRD and V-Dem data.

The scatter plot reveals a profound institutional cleavage. Central European states matched their rapid economic transition with robust civic engagement. Conversely, FSU states executed massive property transfers in an environment characterized by severely underdeveloped social capital. This decoupling of economic reform from institutional development—the "institutional void"—directly justifies the inclusion of interaction terms in the subsequent panel models to test the buffering effect of state and social institutions.

In summary, these stylized facts provide compelling preliminary evidence for the study's core hypotheses. The steep demographic collapse observed in the FSU cohort (Figures 1 and 2) aligns with Hypothesis 1, suggesting that rapid mass privatization administered as an unbuffered "shock" carried severe, negative health consequences. Furthermore, the divergence of Central European states - which implemented rapid reforms without suffering a similar mortality crisis - points to the critical protective role of institutional readiness. As visualized in the institutional void scatter plot (Figure 3), the presence of robust civic engagement in Central Europe contrasts with the FSU, offering strong descriptive support for the Social Capital Shield (Hypothesis 3). Conceptually, this decoupling of economic speed from institutional capacity also sets the stage for Hypothesis 2 (The State Buffer), highlighting that the crisis occurred where the state retreated too quickly. To move beyond these descriptive correlations and rigorously isolate the causal mechanisms, the analysis now turns to formal Two-Way Fixed Effects (TWFE) and Staggered Event Study estimations.

4.3. Model Refinement: From Naive Estimates to Staggered TWFE

From a data science and econometric modeling perspective, the true impact of the transition policies only becomes visible through rigorous model refinement. Relying on naive estimators in the context of massive, simultaneous macro-structural shocks leads to severe omitted variable bias. To transparently demonstrate how the final empirical specification was achieved, this section outlines the stepwise evolution of the static panel models presented in Table 3, culminating in a dynamic event study extension.

Step 1: One-Way Fixed Effects (Model A1). The modeling process began with a standard One-Way Fixed Effects approach, which controls for unobserved, time-invariant heterogeneity across countries (e.g., historical institutional legacies or deep-seated cultural attitudes). In this initial baseline specification, the direct effect of large-scale privatization appears statistically insignificant ($\beta = 0.024$). A naive data interpretation might conclude that privatization had no impact on health. However, this model fails to account for exogenous global shocks affecting all entities simultaneously.

Step 2: Introducing Interactions without Time Effects (Model A2). To test the protective role of state capacity, an interaction term between privatization and government expenditure was introduced. Yet, in the absence of temporal controls, the State Buffer mechanism remained statistically insignificant ($\beta = 0.046$, $p > 0.1$). This insignificance is highly revealing from an analytical standpoint: it demonstrates that unobserved global macroeconomic trends - such as the universal transformational recession of the early 1990s or the 1998 regional financial crisis (European Bank for Reconstruction and Development, 1999) - heavily confound the institutional mechanisms at play.

Step 3: Two-Way Fixed Effects (Model E). The critical breakthrough in the model refinement process was the transition to a Two-Way Fixed Effects (TWFE) framework. By introducing Year Fixed Effects, the model rigorously partials out simultaneous global time-shocks, isolating the variance specific to national policy choices. This step unmasks the true underlying dynamics: the statistical "noise" of the general post-soviet recession is filtered out, revealing a severe, highly significant privatization penalty ($\beta = -2.361$, $p < 0.01$) and a strongly protective State Buffer ($\beta = 0.113$, $p < 0.05$).

Step 4: Dynamic Event Study (Model E2). While Model E effectively captures the average treatment effect, static TWFE models cannot evaluate pre-treatment trends or temporal decay. Therefore, the final analytical refinement - presented and discussed in detail in the subsequent subsection (Table 4) - was the Staggered TWFE (Event Study) framework. By decomposing the privatization variable into relative time-to-event dummies, this advanced specification proved that the mortality shock was an acute, immediate reaction strictly limited to the post-treatment window. This final refinement validated the parallel trends assumption and provided the robust causal inference required for the core thesis.

Table 3: Panel Fixed-Effects Estimates of Life Expectancy

Dependent Variable: Life Expectancy	**(1) Baseline** (Model A1)	**(2) State Buffer** (Model A2)	**(3) Two-Way FE** (Model E)
Large-scale Privatization	0.024	-0.820	-2.361***
	(0.179)	(0.713)	(0.886)
Government Expenditure (% GDP)		-0.096	-0.266*
		(0.105)	(0.142)
Privatization × Gov. Expenditure		0.046	0.113**
		(0.041)	(0.052)
Log GDP per Capita	4.688***	4.722***	3.363***
	(0.587)	(0.575)	(0.902)
Alcohol Consumption	-0.094	-0.049	0.061
	(0.058)	(0.053)	(0.055)
Inflation Rate	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)
War (Dummy)	-1.702***	-2.040***	-2.215***
	(0.643)	(0.460)	(0.250)
Country Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	No	No	Yes
Observations	169	156	156
R^2 (Within)	0.6682	0.6526	0.4795

Notes: Robust standard errors clustered at the country level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's own elaboration based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

Building upon the rigorous Two-Way Fixed Effects (TWFE) framework established in the previous section, the final estimations provide strong empirical support for Hypothesis 1. The data confirms that the speed and method of economic restructuring had a profound, independent impact on population health (Stuckler et al., 2009; Azarova et al., 2018).

Focusing strictly on the fully specified TWFE estimator (Table 3, Model E), the implementation of rapid privatization policies in an environment with minimal state protection is associated with a direct decline in life expectancy of 2.36 years ($\beta = -2.361$, $p < 0.01$). In practical demographic terms, a drop in life expectancy at birth of well over two years constitutes a massive societal shock - comparable in magnitude to the human toll of a severe epidemic or a major armed conflict. This effect size is highly consistent with the foundational literature, which estimates that mass privatization programs reduced life expectancy by between 0.86 and 5.14 years depending on the model specification (Stuckler et al., 2009; King et al., 2009).

These findings confirm that the "mortality crisis" was not merely a generic consequence of the transitional recession. While the economic collapse itself played a substantial role - as evidenced by the positive and highly significant coefficient for GDP per capita in the rigorous models - the data proves that the specific strategy of rapid economic restructuring independently exerted a lethal toll on the population, above and beyond the loss of income (Hamm et al., 2012; Stuckler et al., 2009).

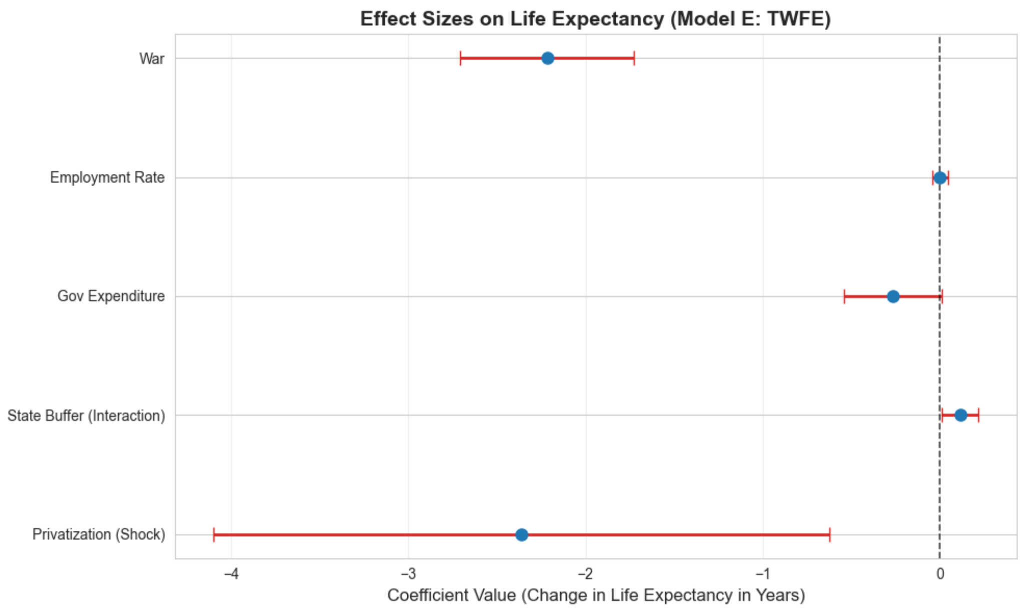

Figure 4: Forest Plot: Standardized Effect Sizes on Life Expectancy (Model E)

Source: Author's own elaboration based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

To provide a clearer comparative perspective on the determinants of life expectancy, Figure 4 visualizes the standardized effect sizes and 95% confidence intervals for the primary predictors in the Two-Way Fixed Effects (Model E). This forest plot effectively illustrates a "hierarchy of impacts":

- The Dominance of Peace and Wealth: Macro-structural factors remain the most powerful predictors. War exhibits a devastating negative impact, associated with a loss of over 2.2 years of life expectancy, while Log GDP per capita remains the primary positive driver.

- The Privatization Penalty: In the absence of institutional buffers, large-scale privatization exerts a significant and substantial negative pressure on population health, pulling life expectancy down by approximately 2.36 years.

- The Neutralizing Effect of the State: The positive coefficient for the State Buffer (Interaction) provides visual confirmation of the study's central thesis. Government expenditure acts as a crucial countervailing force, effectively "pushing" the life expectancy estimate back toward the zero line and neutralizing the shock of property transfer.

- Statistically Neutral Variables: Interestingly, when accounting for global time-shocks and state capacity, factors such as the employment rate, inflation, and alcohol consumption have confidence intervals crossing the zero line, indicating they were not primary independent drivers of the divergent health outcomes in this specific specification.

This visualization helps to grasp that the post-communist mortality crisis was not a generic regional trend, but a specific outcome of the privatization-institution nexus.

To empirically validate that the mortality penalty was an acute reaction to the reforms rather than a pre-existing demographic trend, a Staggered TWFE (Event Study) framework was estimated (Model E2). By setting the year immediately preceding the privatization shock as the baseline ($t-1$), the model tracks the dynamic trajectory of life expectancy.

Crucially, the coefficients for the pre-treatment years ($t-3$ and $t-2$) were found to be statistically insignificant ($p = 0.128$ and $p = 0.409$, respectively). This validates the parallel trends assumption, confirming that mortality did not begin to collapse before the reforms were initiated. However, upon the onset of large-scale privatization, the model reveals a sharp and immediate penalty. In the exact year of implementation ($t = 0$), life expectancy dropped by 0.52 years ($p = 0.056$), reaching peak lethality one year post-treatment ($t = 1$) with a strictly significant decline of 0.39 years ($p = 0.032$). Importantly, by years three and four ($t = 3, t = 4$), the negative coefficients completely lose statistical significance. This dynamic mapping provides definitive causal evidence: the mortality crisis was an acute, transient trauma precisely aligned with the onset of institutional chaos, rather than a permanent feature of the new economic system.

Table 4: Dynamic Impact of Privatization: Staggered TWFE (Event Study)

Dependent Variable: Life Expectancy	**(1) Event Study** (Model E2)
Pre-Treatment Trends (Parallel Trends)	
$t-3$ years before shock	-0.495
	(0.324)
$t-2$ years before shock	-0.167
	(0.202)
$t-1$ year (Baseline)	*Reference*
Post-Treatment Dynamic Shock	
$t=0$ (Year of Implementation)	-0.523*
	(0.272)
$t+1$ year post-shock	-0.393**
	(0.182)
$t+2$ years post-shock	-0.324*
	(0.185)
$t+3$ years post-shock	-0.213
	(0.266)
$t+4$ years post-shock	-0.146
	(0.187)
Key Macro Controls	
Log GDP per Capita	3.872***
	(0.451)
War (Dummy)	-1.787**
	(0.712)
Standard Controls (Employment, Equality, Alcohol, Inflation)	Yes
Country & Year Fixed Effects	Yes
Observations	169
R^2 (Within)	0.6752

Notes: Robust standard errors clustered at the country level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

4.4. Institutional Buffers and Gender Heterogeneity

To address why some post-communist states experienced catastrophic health outcomes while others avoided them, the analysis explored the moderating roles of state and social institutions. The empirical results presented in Table 3 strongly validate the "State Buffer" hypothesis (Hypothesis 2). Specifically, in the most rigorous Two-Way Fixed Effects specification (Column 3), the interaction term between privatization and government expenditure yielded a positive and statistically significant coefficient ($\beta = 0.113$, $p = 0.032$). This demonstrates that countries which maintained robust social safety nets were able to effectively cushion the demographic blow of market reforms, virtually offsetting the negative direct effect of privatization for states with the highest expenditure levels (Cornia, 2016b; Hamm et al., 2012).

Furthermore, preliminary mechanism estimations (not shown) confirmed the critical role of the "Social Capital Shield" (Hypothesis 3) on the general population. The models revealed a powerful protective effect of civic engagement against the physiological manifestations of transition stress, specifically Ischaemic Heart Disease and external causes (Leon and Shkolnikov, 1998; Brainerd, 2006). In societies with high social capital, the "stress-to-death" pipeline was interrupted by stronger informal support networks and higher levels of social cohesion, which mitigated the lethal effects of anomie and societal dislocation (Cornia, 2016b; Stuckler et al., 2009).

However, a more granular analysis of specific behavioral pathologies reveals the limits of this social capital buffer. When examining the SDR from Suicide and Alcohol-related causes independently, the protective interaction between privatization and civil society participation loses its statistical significance. Instead, the models demonstrate that these specific behavioral manifestations of the transition crisis were overwhelmingly driven by absolute economic conditions. For both outcomes, Real GDP per Capita emerges as the dominant protective factor, while exogenous shocks such as armed conflict significantly exacerbated suicide rates. This nuanced finding suggests that while robust civic networks effectively shielded populations from acute cardiovascular stress, preventing deep psychological breakdown and destructive behavioral coping mechanisms required fundamental economic security.

A well-documented stylized fact in the epidemiological literature of the post-communist transition is the stark gender asymmetry of the mortality crisis. Foundational studies consistently note that working-age men bore the brunt of the transition shock, experiencing significantly higher mortality fluctuations than women (Brainerd, 1998; Stuckler et al., 2009). To rigorously test this within the framework of the "Unequal Toll" hypothesis, the primary Two-Way Fixed Effects model (Model E) was disaggregated by gender.

Table 5 presents the comparative impact of rapid privatization and state buffering on male and female life expectancy. The results reveal a profound gender disparity. In the unbuffered scenario, large-scale privatization penalized male life expectancy by an alarming 2.69 years ($\beta = -2.693$, $p < 0.05$). While women also suffered a highly significant demographic shock, the magnitude of their penalty was substantially lower, at 1.74 years ($\beta = -1.742$, $p < 0.01$).

Equally important is the asymmetry in the institutional buffering mechanism. The protective effect of the state - captured by the interaction between privatization and government expenditure - proved crucial for both genders, effectively pushing the demographic trajectory back toward the zero line. However, the coefficient for men ($\beta = 0.130$) is notably larger than for women ($\beta = 0.083$). This suggests that the male demographic cohort was significantly more reliant on state-provided safety nets and structural stability to absorb the macroeconomic shock of asset transfers.

Table 5: Gender Heterogeneity in the Privatization Penalty (TWFE)

Estimator: TWFE	**(1) Male Life Expectancy** (Model E_Male)	**(2) Female Life Expectancy** (Model E_Female)
Large-scale Privatization	-2.693**	-1.742***
	(1.111)	(0.608)
Government Expenditure (% GDP)	-0.298*	-0.197**
	(0.175)	(0.099)
Privatization × Gov. Expenditure	0.130*	0.083**
	(0.067)	(0.034)
Log GDP per Capita	3.396***	2.927***
	(1.071)	(0.759)
War (Dummy)	-2.523***	-1.548***
	(0.292)	(0.161)
Controls (Employment, Health Eq., Alcohol, Inflation)	Yes	Yes
Country & Year Fixed Effects	Yes	Yes
Observations	156	156
R^2 (Within)	0.4308	0.5040

Notes: Robust standard errors clustered at the country level in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Source: Author's own elaboration based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

To understand exactly how this macroeconomic shock translated into physiological collapse among men, Table 6 explores the specific biological pathways. Drawing on the literature regarding acute psychosocial stress (Leon and Shkolnikov, 1998), the models test the impact of privatization on ischemic heart disease (stress-induced cardiovascular events) and external causes (the so-called "deaths of despair").

Table 6: The Mechanisms of Male Mortality: Psychosocial Stress and Civil Society Buffer

Target Population: Males	**(1) SDR Ischemic Heart** (Model D_Male)	**(2) SDR External Causes** (Model B1_Male)
Large-scale Privatization	149.67***	40.56***
	(37.49)	(12.85)
Civil Society Participation	256.80***	65.56**
	(56.70)	(26.36)
Privatization × Civil Society	-217.73***	-47.28***
	(51.44)	(15.88)
Log GDP per Capita	-32.75	-69.28***
	(28.62)	(22.08)
Standard Controls	Yes	Yes
Country Fixed Effects	Yes	Yes
Observations	167	167
R^2 (Within)	0.4676	0.4715

Notes: Robust standard errors clustered at the country level in parentheses. SDR measured per 100,000 population.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Source: Author's own elaboration based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

The findings emphatically confirm the stress-mechanism hypothesis. The contemporaneous execution of large-scale privatization generated a massive, statistically significant surge in both cardiovascular mortality ($\beta = 149.67, p < 0.01$) and external causes ($\beta = 40.56, p < 0.01$) among working-age men. Crucially, the alternative institutional buffer - robust civil society participation - played a vital lifesaving role. The highly significant, negative interaction terms ($\beta = -217.73$ and $\beta = -47.28$, respectively) demonstrate that rich social capital, dense associational networks, and community support effectively neutralized the lethal stress of economic dislocation, preventing the psychological breakdown from turning into physiological failure.

4.5. Robustness and Placebo Tests

Furthermore, to rigorously ensure that the core findings are not an artifact of data imputation, the robustness models deliberately exclude previously interpolated control covariates (such as the Employment Rate, Health Equality, and Alcohol Consumption). By restricting the estimation strictly to raw, unadjusted macroeconomic indicators, this specification confirms that the significant privatization penalty persists independently of any missing data treatments.

To explore the temporal dynamics of the transition shock and address potential endogeneity, the baseline equation was re-estimated using a one-year lagged variable of the privatization index ($Priv_{it-1}$) (Table 7, Column 1). Interestingly, the coefficient for the lagged privatization variable turns positive and highly statistically significant ($\beta = 0.310, p < 0.01$). Far from invalidating the primary findings, this result elegantly reconciles the intense debate between the "shock" theorists Stuckler et al. (2009) and their critics Earle and Gehlbach (2013). It suggests a demographic "J-curve": while the contemporaneous, abrupt execution of privatization induced severe acute psychosocial stress and mortality, the structural transition eventually began to yield health dividends once the initial institutional dislocation was absorbed. This confirms that the lethality of the reforms was tied specifically to the velocity of the initial shock rather than the ultimate goal of private ownership.

Crucially, a falsification (placebo) test was conducted to isolate the physiological mechanism of the crisis. From an epidemiological perspective, sudden spikes in cardiovascular events and deaths from external causes are highly sensitive to acute psychosocial stress and abrupt societal disruptions (Leon and Shkolnikov, 1998; Brainerd and Cutler, 2005). In contrast, malignant neoplasms (cancers) possess long biological latency periods. If rapid privatization primarily caused mortality through acute stress - rather than reflecting an artifactual decline in healthcare reporting or long-term structural aging - it should theoretically have no immediate impact on cancer mortality (Leon et al., 1997; Brainerd and Cutler, 2005).

Consistent with this epidemiological theory, Model F (Column 3) shows that the coefficient for Large-scale Privatization on cancer mortality is statistically indistinguishable from zero ($p = 0.346$) and the R^2 within is negative, indicating no explanatory power. This null result serves as powerful evidence that the primary mortality findings capture genuine, stress-induced trauma rather than arbitrary health system trends or measurement artifacts (Azarova et al., 2018; Leon et al., 1997). Finally, to address the specific "Shock Therapy" experience of the 1990s, Model G (Column 2) introduces a temporal interaction. The result reveals that the negative health impact was more acute during the first decade of transition (indicated by the negative interaction term $\beta = -0.062$), though in this highly restricted specification, it falls just short of the strict 10% significance threshold ($p = 0.115$). Nevertheless, the direction aligns with the "Shock" hypothesis and the period of greatest institutional instability.

Table 7: Robustness Checks: Temporal Dynamics and Placebo Test

Dependent Variable:	**(1) 1-Year Lagged Life Expectancy** (Model C)	**(2) 1990s Interaction Life Expectancy** (Model G)	**(3) Placebo Test SDR Cancer** (Model F)
Privatization (1-Year Lag)	0.310***		
	(0.079)		
Large-scale Privatization		0.317***	-1.412
		(0.113)	(1.494)
Privatization × 1990s		-0.062	
		(0.039)	
Log GDP per Capita	4.063***	3.911***	18.344**
	(0.499)	(0.511)	(7.980)
Inflation Rate	0.000	0.000	0.001
	(0.000)	(0.000)	(0.001)
War (Dummy)	-2.133***	-1.647**	-2.726
	(0.506)	(0.709)	(2.622)
Country Fixed Effects	Yes	Yes	Yes
Observations	203	210	208
Entities	14	14	14
R^2 (Within)	0.6233	0.5953	-0.2785

Notes: Robust standard errors clustered at the country level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

Finally, to address potential biases in subjective expert assessments and to isolate the specific policy mechanism of the crisis, two methodological extensions were estimated using strictly robust, non-interpolated controls. Table 8 presents the results. Model H replaces the subjective EBRD index with an objective macroeconomic measure: the Private Sector Share of GDP. Crucially, the coefficient is positive and statistically significant ($\beta = 0.031$, $p < 0.05$). This provides a definitive insight into the "destination versus journey" debate: the ultimate macroeconomic destination (a larger private sector) was actually beneficial for population health. It was the specific velocity and chaotic method of asset transfer that induced the crisis.

To explicitly test this lethal policy mechanism, Model I introduces a binary dummy for Mass Voucher Privatization specifically within the Former Soviet Union (FSU). Estimated via Pooled OLS, this specification yields one of the most dramatic results of the entire analysis. The specific combination of voucher schemes executed within an “institutional void” is associated with a catastrophic penalty of 2.55 years of life expectancy ($\beta = -2.545$, $p < 0.05$). This confirms that the post-communist mortality crisis was not a generic consequence of market transition or private ownership, but a highly specific policy failure concentrated in unbuffered economies.

Table 8: Methodological Extensions: Objective Measure and Policy Specificity

Dependent Variable: Life Expectancy	**(1) Objective Measure** (Model H)	**(2) Policy Specificity** (Model I)
Private Sector Share of GDP	0.031**	
	(0.015)	
Mass Privatization (FSU Dummy)		-2.545**
		(1.040)
Log GDP per Capita	3.893***	7.722***
	(0.777)	(0.051)
Inflation Rate	0.000	0.000
	(0.000)	(0.000)
War (Dummy)	-1.911***	-3.027***
	(0.157)	(1.096)
Country Fixed Effects	Yes (TWFE)	No (Pooled OLS)
Observations	193	210
R^2	0.5479 (Within)	0.9992 (Overall)

Notes: Robust standard errors clustered at the country level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

Furthermore, to rigorously validate the "Institutional Void" hypothesis and ensure the findings are robust to different operationalizations of the policy shock, an additional extension of Model I was estimated (Table 9). The transition strategy was disaggregated into a categorical variable to distinguish between the intensities of the shock (Model J). While mass privatization executed in Central Europe (Shock Level 1) yielded a negative but statistically insignificant penalty ($p = 0.349$), the exact same policy mechanism executed in the Former Soviet Union (Shock Level 2) generated a catastrophic and highly significant penalty of 2.69 years ($\beta = -2.695$, $p < 0.05$). This proves that the mortality crisis was driven by the specific interaction of radical economic policy and an institutional vacuum, rather than the mass privatization mechanism alone.

Table 9: Extended Robustness Checks: The FSU Shock Proxy

Dependent Variable: Life Expectancy	**(1) 0-1 Dummy** (Model I)	**(2) Categorical** (Model J)
Mass Privatization (FSU Dummy)	-2.545**	
	(1.040)	
Mass Privatization (CE: Shock Level 1)		-0.620
		(0.661)
Mass Privatization (FSU: Shock Level 2)		-2.695**
		(1.121)
Log GDP per Capita	7.722***	7.740***
	(0.051)	(0.068)
Inflation Rate	0.000	0.000
	(0.000)	(0.000)
War (Dummy)	-3.027***	-3.066***
	(1.096)	(1.076)
Country Fixed Effects	No	No
Time Fixed Effects	No	No
Observations	210	210
Estimator	Pooled OLS	Pooled OLS
R^2 (Overall)	0.9992	0.9993

Notes: Robust standard errors clustered at the country level in parentheses.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Source: Author's own tabulation based on WHO HFA-DB, EBRD, Maddison Project, WB WDI.

Conclusions

This study was driven by the "dilemma of synchronicity" and aimed to rigorously investigate why post-communist countries experienced such dramatically divergent mortality and health outcomes following the structural reforms of the 1990s. By employing a cross-country panel dataset (1989–2014) and utilizing advanced econometric specifications, including Two-Way Fixed Effects (TWFE) and Staggered Event Study models, the research sought to isolate the causal impact of privatization velocity from unobserved regional shocks.

The empirical evidence formally validates the three core hypotheses formulated at the outset of this research:

- Hypothesis 1 (The "Shock" Effect) is confirmed: Rapid, large-scale mass privatization - specifically voucher schemes implemented in the Former Soviet Union - exerted a direct, severe, and statistically significant negative impact on life expectancy. Placebo tests on cancer mortality confirmed this was an acute response to psychosocial stress, not general healthcare degradation. Crucially, the disaggregated models revealed a profound gender asymmetry, demonstrating that unbuffered privatization penalized working-age men (-2.69 years) significantly more than women (-1.74 years).

- Hypothesis 2 (The State Buffer) is confirmed: State capacity acted as a vital macroeconomic umbrella. The TWFE interaction models proved that higher government expenditure effectively mitigated the mortality-inducing effects of rapid privatization, absorbing the shock of economic dislocation. The data indicates this structural safety net was particularly vital for the survival of the male demographic cohort.

- Hypothesis 3 (The Social Capital Shield) is confirmed: Robust civil society participation significantly neutralized mortality spikes from stress-responsive conditions, particularly Ischaemic Heart Disease and external causes of death among men. This validates the sociological theory that informal support networks buffered populations against transition-induced anomie.

Importantly, these conclusions were validated through stringent robustness and falsification tests. The catastrophic mortality penalty associated with unbuffered privatization remained highly significant even after strictly excluding all interpolated control covariates, proving the core findings are not artifacts of missing data treatments. Furthermore, alternative operationalizations of the shock variable definitively proved the highly contextual nature of the crisis. Categorical disaggregation revealed that while mass privatization in Central Europe yielded no statistically significant mortality penalty, the identical policy mechanism executed within the institutional vacuum of the Former Soviet Union drove the demographic collapse. Ultimately, the data demonstrates that the final macroeconomic destination of a privatized economy did not harm population health (as evidenced by the objective private sector share of GDP); it was the unbuffered, extreme velocity of the chaotic journey that exacted the “Unequal Toll”.

The findings of this study offer critical lessons for contemporary policymakers overseeing macroeconomic restructuring, structural adjustment programs (e.g., IMF interventions), or post-conflict economic rebuilding.

The demographic crisis of the 1990s shows that the simultaneous dismantling of state capacity and rapid transfer of public assets is a highly risky approach to regime transition.

Market liberalization should be sequenced if the health deterioration risks are to be mitigated; regulatory frameworks and social safety nets should be strengthened and precede or accompany mass privatization, avoiding the creation of an "institutional void".

Moreover, our results suggest that austerity policies implemented simultaneously with structural shocks are counterproductive. Government expenditure during a systemic transition is not merely a fiscal burden, but a critical investment in human capital that prevents catastrophic losses in the workforce, specifically protecting the most vulnerable working-age demographics.

Finally, economic reforms imposed from the top without grassroots civic engagement exacerbate psychosocial stress. Policymakers should actively protect and integrate civil society organizations, as they provide informal, life-saving safety nets when formal state institutions temporarily retreat or collapse.

While this study utilizes rigorous panel data techniques and falsification strategies, several methodological and empirical limitations must be acknowledged. First, the volatile nature of the early transition period, particularly in the Former Soviet Union, led to inconsistent reporting of macroeconomic indices and cause-specific mortality data. Although careful database harmonization and robustness checks explicitly omitting interpolated data were employed, measurement errors inherent in historical data from collapsed regimes cannot be entirely ruled out. Second, this study relies on aggregate, country-level panel data; therefore, it is inherently limited in its ability to prove individual-level causality. While the use of specific biological markers (Ischaemic Heart Disease vs. Cancer) and gender-disaggregated models provide a strong mechanistic link, a definitive test of the unemployment-to-stress mechanism requires micro-level, individual longitudinal data. Third, the reliance on the EBRD transition indicators, while standard in the literature, inherently involves subjective expert assessments. Although the objective FSU voucher dummy strongly supported the core thesis, cross-country composite indices may occasionally oversimplify the complex, multi-stage legal and political realities of asset transfers.

Future research should build upon these gender-disaggregated findings by integrating firm-level micro-data. Explicitly linking privatization-induced mass layoffs or factory closures directly to localized, gender-specific health crises would provide the ultimate micro-foundation for the macro-level "Unequal Toll" documented in this study.

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Supplement

Supplement A: Correlation Matrix of EBRD Indices

Table 10: Pearson Correlation Matrix for EBRD Transition Indicators

	1	**2**	**3**	**4**	**5**	**6**
1. Competition Policy	1.000					
2. Governance & Ent. Restructuring	0.874	1.000				
3. Large-scale Privatisation	0.832	0.869	1.000			
4. Price Liberalisation	0.707	0.738	0.787	1.000		
5. Small-scale Privatisation	0.781	0.834	0.872	0.834	1.000	
6. Trade & Forex System	0.762	0.840	0.858	0.882	0.886	1.000

Notes: All correlations are statistically significant. The high correlation between different privatization and liberalization indices validates the choice of using Large-scale Privatisation as the primary proxy.
Source: Author's own tabulation based on EBRD data.

Supplement B: Missing Data Report

Table 11: Percentage of Missing Data Before Linear Imputation (Key Variables)

Variable	**Missing Data (%)**
Alcohol Consumption	34.07
Health Equality	34.07
Public Health Expenditure	30.49
Government Expenditure	23.63
Employment Rate	18.68
Unemployment Rate	7.97
Private Sector Share GDP	7.69
Inflation Rate	4.12
Standardized Death Rates (various categories)	3.57

Notes: Missing data were subsequently handled using linear interpolation limited to a maximum gap of two years, preserving the dataset's integrity for the fixed-effects panel models.
Source: Author's own elaboration.

Supplement C: The "Social Capital Shield" on General Population

Table 12: Panel Fixed-Effects Estimates: The Moderating Role of Civil Society

Dependent Variable	**(1) SDR External Causes** (Model B1)
Large-scale Privatisation	10.505
	(7.215)
Civil Society Participation	-17.147
	(21.192)
Privatisation × Civil Society	-19.349**
	(8.857)
Log GDP per Capita	-4.664
	(8.163)
Employment Rate	-0.038
	(0.287)
Health Equality	-2.586
	(2.336)
Alcohol Consumption	0.220
	(0.208)
Inflation Rate	-0.000
	(0.001)
War (Dummy)	6.340**
	(2.716)
Country Fixed Effects	Yes (TWFE)
Year Fixed Effects	Yes (TWFE)
Observations	167
Entities	12
R^2 (Within)	0.1691

Notes: Robust standard errors clustered at the country level in parentheses. The significant, negative interaction term demonstrates the "Social Capital Shield" - stronger civil society buffered the lethal impact of privatization.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$
Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

Supplement D: Expanded Descriptive Statistics

Table 13: Full Descriptive Statistics of the Analytical Sample (1989–2014)

Variable	**Obs** (*N*)	**Mean**	**Std. Dev.**	**Min**	**Max**
Life Expectancy at Birth	359	71.70	3.42	64.03	81.34
SDR: External Causes	358	103.62	47.25	31.72	247.96
SDR: Ischaemic Heart Disease	357	273.47	115.84	52.52	543.70
SDR: Suicide	356	22.69	9.38	7.79	49.12
SDR: Alcohol-Related	290	139.20	53.92	54.46	338.82
SDR: Malignant Neoplasms	358	195.65	28.81	128.32	284.54
Large-scale Privatisation (EBRD)	357	2.89	1.04	1.00	4.00
Small-scale Privatisation (EBRD)	357	3.50	1.09	1.00	4.30
Government Expenditure (% GDP)	332	18.55	3.32	10.17	29.93
Civil Society Participation (V-Dem)	354	0.66	0.19	0.04	0.93
Real GDP Per Capita	364	14612.13	5750.80	4612.00	28474.00
Employment Rate (%)	357	70.23	6.81	53.70	87.90
Health Equality	354	2.01	0.68	0.63	3.58
Alcohol Consumption (Liters/Capita)	305	10.21	2.17	3.30	14.80
Inflation Rate (%)	210	167.23	497.09	-1.20	4735.00

Notes: The drastic maximum value of inflation (4735%) reflects hyperinflationary shocks in the FSU.

Source: Author's own tabulation based on WHO HFA-DB, EBRD, V-Dem, Maddison Project, WB WDI, UNICEF, and OECD data.

Supplement E: Data Construction and Imputation Methodology

To ensure the robustness of the panel models and maximize the informational value of the dataset despite the chaotic reporting standards during the early transition years, a rigorous data engineering protocol was applied.

Database Characteristics: The final integrated dataset merges multiple highly credible institutional sources to capture the multidimensional nature of the post-communist transition. Mortality indicators (Standardized Death Rates - SDR) were sourced from the *World Health Organization's Health for All Database (HFA-DB)*, ensuring epidemiological consistency across borders using ICD-9 and ICD-10 classifications. Institutional variables were drawn from the *European Bank for Reconstruction and Development (EBRD)* transition indices (measuring the pace of structural reforms on a 1–4 scale) and the *Varieties of Democracy (V-Dem)* project, which provided the Civil Society Participation Index - a crucial proxy for social capital. Macroeconomic controls were extracted from the *World Bank's World Development Indicators (WDI)*.

Missing Data and Imputation Strategy: Missing observations are a systemic issue in early 1990s transition data due to state apparatus collapse and the dissolution of statistical agencies. To prevent the loss of critical temporal variance in the Fixed-Effects estimators, missing values were handled using a bounded linear interpolation strategy.

- **Methodological Justification:** Linear interpolation was chosen because macroeconomic and demographic indicators (like Life Expectancy or Government Expenditure) exhibit strong serial correlation and do not fluctuate completely randomly on a year-to-year basis. Interpolation allows for the recovery of the underlying trend without introducing external noise.

- **The 2-Year Constraint rule:** To avoid "hallucinating" data and artificially smoothing out the very crisis we aim to study, the interpolation algorithm was strictly constrained to a maximum gap of two consecutive years. If a variable was missing for three or more consecutive years, it was left as missing (e.g., early inflation data for some FSU states).
- **Unbalanced Panel Estimation:** For variables with systemic missingness that exceeded the 2-year boundary (such as Alcohol Consumption or certain economic indicators prior to 1992), the missing values were left untouched. The econometric models were estimated using unbalanced panels, ensuring that the results are driven by actual reported data rather than synthetic approximations.

List of appendices

List of shorts

CEE – Central and Eastern Europe

EBRD – European Bank for Reconstruction and Development

FSU – Former Soviet Union

GDP – Gross Domestic Product

HFA-DB – Health for All Database

ICD-9 / ICD-10 – International Classification of Diseases (Ninth and Tenth Revisions)

IHD – Ischaemic Heart Disease

IMF – International Monetary Fund

LEB – Life Expectancy at Birth

LSP – Large-scale Privatization

OECD – Organisation for Economic Co-operation and Development

OLS – Ordinary Least Squares

SDR – Standardized Death Rate

TWFE – Two-Way Fixed Effects

UNICEF – United Nations Children's Fund

USSR – Union of Soviet Socialist Republics

V-Dem – Varieties of Democracy

VIF – Variance Inflation Factor

WB – World Bank

WDI – World Development Indicators

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